

Lunar Lander – Physics & Controls Design

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Overview & design goals

A 3D browser game (HTML, WebGL, JavaScript) where the player flies a single rocket from lunar orbit to a safe touchdown. The core challenge is the contrast between a powerful but sluggish main engine and eight weak but instant side thrusters: the player must plan ahead with the engine and correct with the thrusters.

Design goals:

- **Believable physics, not a full simulator.** Real lunar gravity and rocket equation; simplifications where they don't change the feel (no roll, rigid body, no fuel slosh).
- **Readable state.** The player always sees altitude, velocity, attitude and the target, so a crash is their mistake, not a surprise.
- **Scales with missions.** The same model supports a short landing from 1.5 km and a full descent from orbit; missions change only the body, start state, fuel and goals.
- **Optional assists.** Stability assist and prediction aids can be switched on for easy mode and off for hard mode.

World model: coordinates, gravity, scale

The simulation uses one Moon-centred Cartesian frame in SI units (metres, seconds, kilograms). Gravity points to the Moon's centre, so orbit, descent and landing all run on the same equations.

Gravity (point mass, no rotation, no atmosphere):

$$\vec{a}_g = -\frac{GM}{|\vec{r}|^3}\vec{r}, \quad \text{GM} = 4.9048695 \times 10^{12} \text{ m}^3/\text{s}^2, \quad R = 1,737,400 \text{ m}$$

At the surface this gives $g \approx 1.62 \text{ m/s}^2$. A circular orbit at 100 km altitude moves at about 1633 m/s and takes about 118 minutes.

Local frame for the player. All HUD values are shown in a local frame at the landing site: x = east, y = north, z = up (altitude). The landing target sits at local (0, 0). Near the surface the Moon's curvature is small, so terrain can be a flat heightmap tile placed on the sphere.

Precision. JavaScript numbers are 64-bit, so physics stays in Moon-centred coordinates. WebGL is 32-bit, so rendering uses a floating origin: every object is drawn relative to the rocket (or camera) to avoid jitter.

Altitude has two meanings, both shown: altitude above the reference sphere (for orbit) and height above ground (AGL) from the terrain under the rocket (for landing). Landing logic always uses AGL.

Rocket model: mass, main engine, side thrusters

The rocket is a rigid cylinder with one gimbal-free main engine on its axis and eight side thrusters in two rings of four, 90° apart: one ring near the top, one near the bottom. All concrete numbers are in the tuning table at the end.

State per physics step: position $\mathbf{r}$, velocity $\mathbf{v}$, orientation $\mathbf{q}$ (quaternion), angular velocity $\boldsymbol{\omega}$, fuel mass, RCS propellant mass, actual engine throttle.

Mass and fuel. Total mass $m = \text{dry mass} + \text{fuel} + \text{RCS propellant}$. Fuel burns according to the rocket equation, so the rocket gets lighter and more responsive as it descends:

$$\dot{m} = -\frac{F}{I_{sp} g_0}, \quad \Delta v = I_{sp} g_0 \ln \frac{m_0}{m_1}, \quad g_0 = 9.80665 \text{ m/s}^2$$

Main engine (slow). Thrust acts along the rocket's long axis through the centre of mass, so it pushes but never rotates. The player sets a commanded throttle; the actual throttle follows it with a first-order lag plus a rate limit:

$$\dot{\theta}_{\text{act}} = \text{clamp} \left(\frac{\theta_{\text{cmd}} - \theta_{\text{act}}}{\tau}, -k, k \right), \quad F_{\text{main}} = \theta_{\text{act}} F_{\text{max}}$$

- Throttle range 10–100 %; a command below 10 % means engine off (a real deep-throttling limit and a nice tactical decision).
- Restarting from off has an ignition delay of about 0.5 s before thrust builds.
- No fuel → no thrust, regardless of command.

Side thrusters (fast). Eight on/off thrusters in two rings (top and bottom), each ring with +X, −X, +Y, −Y in body frame, at lever arm L_{top} above and L_{bot} below the centre of mass. Each reaches full force in about 50 ms. How the two rings fire together decides the effect:

1. **Rotate** (pitch / yaw): top and bottom thrusters push in opposite directions. Forces cancel and torques add, so the rocket tilts without drifting sideways.
2. **Translate** (lateral adjustment): top and bottom push in the same direction. Torques cancel when $L_{\text{top}} = L_{\text{bot}}$, and the whole rocket slides sideways at $2F/m$ — used for final lateral nudges over the target. If the centre of mass has moved (hard mode), a small residual torque remains for the player or SAS to correct.

Roll is not modelled: the rocket is symmetric, so roll doesn't affect flight. Angular acceleration is τ / I , with the moment of inertia of a solid cylinder recomputed from the current mass:

$$I = \frac{m(3r^2 + h^2)}{12}$$

Side thrusters use a separate small propellant tank. Running it dry leaves the player unable to change attitude, which makes RCS use a resource worth managing.

Centre of mass depends on difficulty.

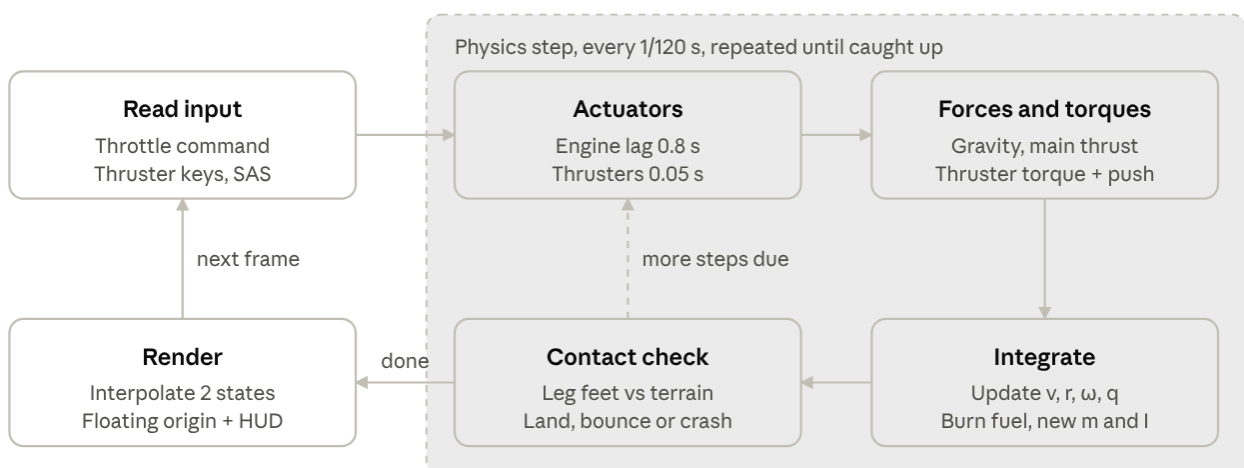
- **Easy mode:** centre of mass fixed at the geometric centre; $L_{\text{top}} = L_{\text{bot}}$ always, so rotate and translate stay perfectly clean.
- **Hard mode:** realistic centre of mass, recomputed every step as the mass-weighted average of dry structure, main fuel (tank low in the rocket) and RCS propellant. As fuel burns the centre of mass climbs: L_{top} shrinks, L_{bot} grows, rotation authority changes and translation picks up a small torque. Moment of inertia is taken about the current centre of mass (parallel-axis theorem).
- The centre of mass only moves along the axis, so main-engine thrust still never rotates the rocket.
- Its position is shown in the auxiliary vehicle view (see Instruments / HUD).

Simulation loop & integration

Physics runs at a fixed 1/120 s step, independent of the display frame rate, so the game behaves the same on a 60 Hz laptop and a 144 Hz monitor.

Each frame reads input, runs fixed physics steps, then draws

Physics at a fixed 1/120 s; rendering at the display rate



game loop · fixed physics step inside each display frame

Each frame adds elapsed time to an accumulator and runs physics steps until it is used up; rendering interpolates between the last two states.

- **Integrator:** semi-implicit Euler (update velocity first, then position with the new velocity). It's stable for orbits at this step size and trivial to write. Quaternion is integrated from ω and re-normalised every step.
- **Spiral guard:** at most 8 steps per frame; if the tab was hidden or the frame was very slow, the extra time is dropped (the game effectively pauses).
- **Time warp:** multiplies the time added to the accumulator ($\times 10$, $\times 100$), allowed only in orbit with the engine off; above $\times 10$, use larger steps or analytic orbit propagation.

Controls

The player sets a throttle target and taps thrusters; the physics does the rest. Keyboard is the baseline, gamepad optional.

Action	Keyboard	Gamepad	Behaviour
Throttle up / down	Shift / Ctrl	Right / left trigger	Changes the commanded throttle by 50 %/s while held; the actual throttle follows with lag
Full throttle	Z	—	Commanded throttle to 100 %
Engine cut	X	B	Commanded throttle to 0 (engine off)
Pitch forward / back	W / S	Left stick up / down	Fires the opposing top/bottom pair while held (pure rotation)
Yaw left / right	A / D	Left stick left / right	Fires the opposing top/bottom pair while held (pure rotation)
Translate forward / back / left / right	I / K / J / L	D-pad	Fires top and bottom thrusters in the same direction while held: slides the rocket sideways without tilting
Stability assist	T	Y	Toggle: automatically fires thrusters to stop rotation

Action	Keyboard	Gamepad	Behaviour
Camera	C to switch view; mouse drag orbits, wheel zooms	Right stick	Several selectable views; each has its own mouse orbit and zoom
Time warp	. / ,	—	Only during coasting in orbit, engine off

Control frame. Side thrusters act in the rocket's body frame, but in 3D a chase camera can swing around, making W mean different things on screen. Default: thruster keys are mapped relative to the camera's horizontal heading, so W always tilts the rocket away from the camera. A "raw body frame" option is available for hard mode.

Cameras. Full 3D from mission 1. During flight the player chooses between several camera views (e.g. chase, free orbit around the rocket, top-down over the target, surface view from the landing site). Every view supports mouse orbit (drag) and zoom (wheel), and remembers its own orbit and zoom when the player switches away and back.

Stability assist (SAS). A simple damping controller: if $|\omega|$ exceeds a small threshold and no thruster key is held, fire the thrusters that oppose ω . It costs RCS propellant and only stops rotation; it never holds a direction. A later upgrade could add "hold retrograde" or "hold vertical" modes.

Why the split matters. The engine's 0.8 s lag means a player who waits for the ground to come close and then hits full throttle is too late. Good play is to set throttle early, then fine-tune attitude with the fast thrusters.

Instruments / HUD

The HUD shows everything the player needs to land without guessing, grouped into five panels plus an auxiliary vehicle view.

Panel	Values shown
Height	Altitude above reference (km in orbit), height above ground AGL (m), time to impact at current vertical speed
Velocity	Vertical speed (m/s, colour: green < 2, yellow < 5, red), horizontal speed and direction arrow, total speed

Panel	Values shown
Attitude	Tilt from vertical (°), tilt direction, angular rate (°/s), small attitude ball showing rocket axis vs. local vertical and velocity vector
Propulsion	Commanded vs. actual throttle (two bars, so the lag is visible), main fuel (kg and %), RCS propellant, remaining Δv (m/s), thrust-to-weight ratio
Navigation	Target distance and bearing, relative x/y position (m), a marker on the ground under the rocket and at the target

Auxiliary vehicle view. A small side-view schematic of the rocket: main and RCS tank fill levels, a centre-of-mass marker with lever arms L_{top} / L_{bot} , and an arrow on each thruster while it fires. In hard mode the marker visibly climbs as fuel burns; in easy mode it stays put. It can be docked in a corner or hidden.

Prediction aids (easy mode, toggleable):

- Projected ground-impact point if the engine stays as it is.
- "Burn now" indicator: the height at which a full-throttle burn (including lag) would stop the descent, compared with the current AGL.
- Velocity vector marker (prograde / retrograde) on the attitude ball, as in real spacecraft, so the player can point the engine against their motion.

Landing & crash rules

Touchdown is judged at the moment the first landing leg touches the ground. The rocket has four legs; each foot is a contact point checked against the terrain height every physics step.

Check	Perfect	Safe	Crash
Vertical speed	< 1.0 m/s	< 2.5 m/s	≥ 2.5 m/s
Horizontal speed	< 0.5 m/s	< 1.5 m/s	≥ 1.5 m/s (tips over)
Tilt from vertical	< 3°	< 10°	$\geq 10^\circ$
Angular rate	< 2 °/s	< 5 °/s	≥ 5 °/s
Ground slope under legs	< 4°	< 10°	$\geq 10^\circ$ (tips over)

- Any body part (not a leg) touching terrain is an instant crash.
- After touchdown the rocket must stay upright for 3 s with the engine off to count as landed (catches bounces and tip-overs).
- Simple leg model: a stiff spring-damper per foot so small bounces look right; no need for a full physics engine.
- Running out of fuel isn't a failure by itself; it just usually leads to a crash.

Score (for later missions): landing grade + remaining fuel + distance to target + time bonus.

Missions

Missions differ only in body, start state, fuel load and goal, so the physics code never changes. A real descent from 100 km orbit lasts over an hour (mostly coasting) and needs about 2 000 m/s of Δv , so the first missions start close to the ground and the full orbit descent comes later.

#	Mission	Start state	Fuel	Goal
1	First touchdown	1 500 m AGL, falling 20 m/s, drifting 30 m/s, upright	500 kg	Land safely on any flat area
2	Pinpoint	3 000 m AGL, 2 km from target, 60 m/s horizontal	700 kg	Land within 50 m (bronze), 20 m (silver), 5 m (gold)
3	Braking burn	15 km altitude, about 1 690 m/s horizontal, roughly 450 km before the site	2 500 kg	Kill orbital speed and land near the target
4	From orbit	100 km circular orbit, engine off	2 600 kg	Deorbit burn (~20 m/s retrograde), coast with time warp, then powered descent and pinpoint landing
5+	Challenges	Varies	Varies	Boulder fields, landing in shadow, limited RCS, engine-out recovery, hopping between two sites

Mission 1 is the tutorial: stability assist on, prediction aids on, generous fuel (about 540 m/s Δv for a job needing roughly 150 m/s). Later missions tighten fuel and turn aids off.

Bodies and terrain. The first missions (1-2) take place on a procedurally generated small moon: radius and GM are chosen so gravity and orbital speeds stay playable, and its heightmap comes from a seed (craters, boulders, flat landing zones), so attempts can vary. Higher missions (3+) use the real Moon at real scale: a coarse global elevation model, plus high-resolution terrain patches around each target modelled on real lunar data (e.g. LRO elevation models of the landing site).

Tuning table & open questions

Starting values, loosely based on a small Apollo-style lander. Everything here is a constant in one config file so it can be tuned by playtesting.

Parameter	Value	Notes
Dry mass	2 000 kg	Without fuel
Rocket size	6 m tall, 1.5 m radius	Used for moment of inertia
Main engine max thrust	12 kN	Thrust-to-weight on the Moon ≈ 2.1 at 3 500 kg, ≈ 1.6 at 4 560 kg
Main engine Isp	311 s	Fuel flow ≈ 3.9 kg/s at full throttle
Throttle range	10–100 %	Below 10 % = off
Throttle lag τ	0.8 s	The "slow response"
Throttle rate limit k	60 %/s	Caps how fast thrust can change
Ignition delay	0.5 s	From engine off only
Side thruster force	400 N each	8 thrusters, two rings of 4
Side thruster lever arm	3 m	Top ring above, bottom ring below the centre of mass; varies with fuel in hard mode
Side thruster response	0.05 s	The "fast response"
Side thruster Isp	220 s	≈ 0.19 kg/s per thruster
RCS propellant	60 kg	≈ 320 thruster-seconds
Angular acceleration	$\approx 11^\circ/\text{s}^2$ at 3 500 kg (one pair)	Faster as fuel drains
Lateral acceleration (translate)	≈ 0.23 m/s ² at 3 500 kg	2×400 N; grows as fuel drains

Parameter	Value	Notes
Physics step	1/120 s	Fixed
Lunar gravity at surface	1.62 m/s ²	From GM and R

Open questions

- ☒ ~~Keep full 3D (x, y, altitude) from mission 1, or start with a 2D side view and add the third axis later?~~
- ☒ ~~Should side thrusters only rotate the rocket, or also have a translate mode (fire opposite pairs) for final lateral nudges?~~
- ☒ ~~Shift the centre of mass as fuel drains (more realistic, harder to tune)?~~
- ☒ ~~Real scale Moon for missions 3–4, or a scaled down Moon to shorten the orbit phase?~~
- ☒ ~~One shared fuel tank or separate main and RCS tanks (decided: separate)?~~
- ☒ ~~Terrain source: procedural heightmap or real lunar elevation data around a known site?~~

Decided 4 Oct: full 3D from mission 1 with selectable cameras (orbit + zoom each); side thrusters in two rings for rotate and translate; centre of mass fixed in easy mode, realistic in hard mode and shown in the auxiliary view; small procedural moon for early missions, real Moon with real-data terrain patches later. Separate main and RCS tanks.